



National
Semiconductor™

Discrete POWER & Signal
Technologies

NPN General Purpose Amplifiers and Switches

Device No.	Case Style	V _{CEO} (V) Min	V _{CE0} (V) Min	V _{EB0} (V) Min	I _{CBO} (nA) @ V _{CB} (V) Max	h _{FE} @ I _C (mA) Min Max	V _{CE(SAT)} (V) & V _{BE(SAT)} (V) @ I _C (mA) Max Min	C _{ob} (pF) Max	f _T (MHz) @ I _C (mA) Min	t _{on} (ns) Max	NF (dB) Max	Test Conditions	Process No.
2N3390	TO-92 (94)	25	25	5	100 18	400 800 2	4.5	10					10
2N3391A	TO-92 (94)	25	25	5	100 18	250 500 2	4.5	10			5	(Note 5)	10
2N3392	TO-92 (94)	25	25	5	100 18	150 300 2	4.5	10					10
2N3393	TO-92 (94)	25	25	5	100 18	90 180 2	4.5	10					10
2N3415	TO-92 (94)	25	25	5	100 25	180 540 2	4.5		0.3 0.6 1.3 50				10
2N3416	TO-92 (94)	50	50	5	100 25	75 225 2	4.5		0.3 0.6 1.3 50				10
2N3417	TO-92 (94)	50	50	5	100 25	180 540 2	4.5		0.3 0.6 1.3 50				10
2N3704	TO-92 (94)	50	30	5	100 20	100 300 50	2	12	100 50				10
2N5172	TO-92 (94)	25	25	5	100 25	100 500 10	10	10					10
MPS8088	TO-92 (92)	60	60	6	100 60	100 300 1 5 100 10 5 75 100 5	0.3	6	150 10				10
MPSA20	TO-92 (92)		40	4	100 30	40 400 5 10		4	125 5				10